

Range of Motion Values

CMT: CMT-0019 (SR)

MCG Health

Shared Resources

28th Edition

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Introduction

Range of motion (ROM) describes the movement of a joint. This tool describes the normal or anticipated full movement for a specific body joint. This information may be useful to identify that ROM is impaired and if rehabilitation therapy services may help improve the ROM range. Active range of motion (AROM) is defined as unassisted voluntary movement of a joint. Active assistive ROM (AAROM) is used when muscles are weak or joints are painful when moved. The therapist helps the patient, but the patient moves the joint. Passive range of motion (PROM) is defined as joint movements performed by the therapist.(1) Normal degree ROM provides the maximum degrees possible for active joint function in a healthy individual. The therapist will provide ROM for patients requiring AAROM or PROM in their care plan. This quick reference represents the normative values for range of motion, and any ranges for these values reflect the variation in the literature.

Quick Reference Range of Motion - Normative Values

JOINT	MOVEMENT	NORMAL DEGREE ROM
Thumb	Radial abduction	Greater than or equal to 50 degrees
	MCP flexion	Greater than or equal to 50 - 60 degrees
	MCP extension	Greater than or equal to 0 degrees
	IP flexion	Greater than or equal to 80 degrees
	IP extension	Greater than or equal to (+)10 - 20 degrees[A]
Finger	MCP flexion	Greater than or equal to 90 degrees

	MCP extension	Greater than or equal to (+)20 - 45 degrees[A]
	PIP flexion	Greater than or equal to 100 degrees
	PIP extension	Greater than or equal to 0 degrees
	DIP flexion	Greater than or equal to 70 - 90 degrees
	DIP extension	Greater than or equal to 0 degrees
Wrist	Flexion	Greater than or equal to 80 - 90 degrees
	Extension	Greater than or equal to 60 - 70 degrees
	Radial deviation	Greater than or equal to 20 degrees
	Ulnar deviation	Greater than or equal to 30 - 55 degrees
Elbow	Flexion	Greater than or equal to 150 - 160 degrees

Forearm	Extension	Greater than or equal to 0 degrees
	Pronation	Greater than or equal to 80 - 90 degrees
	Supination	Greater than or equal to 90 degrees
Shoulder	Flexion	Greater than or equal to 180 degrees
	Extension	Greater than or equal to 50 - 60 degrees
	Abduction	Greater than or equal to 180 degrees
	Adduction	Greater than or equal to 45 - 50 degrees
	Internal rotation	Greater than or equal to 70 - 90 degrees
	External rotation	Greater than or equal to 90 degrees
Great toe	IP flexion	Greater than or equal to 30 - 45 degrees
	MP extension	Greater than or equal to 50 - 70 degrees

2nd to 5th toes	IP flexion	Greater than or equal to 50 degrees
	MP extension	Greater than or equal to 0 degrees
Ankle	Dorsiflexion	Greater than or equal to 15 - 20 degrees
	Plantarflexion	Greater than or equal to 40 - 50 degrees
Hindfoot	Inversion	Greater than or equal to 20 - 35 degrees
	Eversion	Greater than or equal to 10 - 20 degrees
Knee	Flexion	Greater than or equal to 130 - 150 degrees
	Extension	Greater than or equal to 0 - (+)15 degrees
Hip	Flexion	With knee extension greater than or equal to 90 degrees With knee flexion greater than or equal to 100 - 125 degrees
	Extension	Greater than or equal to 30 degrees

	Abduction	Greater than or equal to 45 degrees
	Adduction	Greater than or equal to 30 degrees
	Internal rotation	Greater than or equal to 40 - 50 degrees
	External rotation	Greater than or equal to 40 - 45 degrees
Cervical	Flexion	Greater than or equal to 45 degrees
	Extension	Greater than or equal to 55 degrees
	Lateral flexion (L/R)	Greater than or equal to 40 degrees
	Rotation (L/R)	Greater than or equal to 70 degrees
Thoracic or lumbar	Flexion	Greater than or equal to 75 degrees
	Extension	Greater than or equal to 30 degrees
	Lateral flexion	Greater than or equal to 35 degrees

Rotation Greater than or equal to 30 degrees

(2)(3)(4)

References

1. Walsh C. Assessment: musculoskeletal system. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:1622-1637. [Context Link 1]

2. Musculoskeletal system. In: Ball JW, Dains JE, Flynn JA, Solomon BS, Stewart RW, editors. Seidel's Guide to Physical Examination. 10th ed. Elsevier; 2023:537-580. [Context Link 1]

3. Normative range of motion of the extremities and spine in adults. In: Reese NB, Bandy WD, Yates C, editors. Joint Range of Motion and Muscle Length Testing. 3rd ed. St. Louis, MO: Elsevier; 2017:535-554. [Context Link 1]

4. Patient/client management. In: Dutton M, editor. Dutton's Orthopaedic Examination, Evaluation, and Intervention. 5th ed. McGraw-Hill Education; 2020:163-213. [Context Link 1]

Footnotes

[A] (+) indicates joint hyperextension. [A in Context Link 1, 2]

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